Uncertain Category-Aware Fusion Network for Hyperspectral and LiDAR Joint Classification

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***Abstract*— The integration of hyperspectral (HS) imagery and light detection and ranging (LiDAR) for land cover classification has become a significant research topic. Numerous existing methods aim to interactively fuse the complementary features of HS and LiDAR to enhance the classification accuracy. However, most existing studies overlook the fact that spectral, spatial, and elevation features of HS and LiDAR possess significant discriminative information for specific categories. The rough and simple interacting or stacking these features may hinder the effective expression of this significant discriminative information. Moreover, existing approaches neglect the shared spatial characteristics between HS and LiDAR. In this article, an uncertain category-aware fusion network (UCAFNet) is proposed to tackle the above challenges. Specifically, we proposed an uncertain category-aware fusion strategy (UCAFS) that dynamically weights the spectral, spatial, and elevation branches based on their respective capabilities in identifying different categories to achieve targeted information aggregation. Moreover, we introduce the spatial information purification module (SIPM) and adaptive weighted fusion module (AWFM), to extract and enhance shared spatial features from HS and LiDAR for effective integration. The experimental results on three public benchmark datasets demonstrate the superior performance of the proposed UCAFNet.**

***Index Terms*— Category-aware, convolutional neural network (CNN), hyperspectral (HS), light detection and ranging (LiDAR).**

I. Introduction

THE rapid advancement of remote sensing imaging technology offers robust technical support for the precise monitoring of the Earth's natural resources [1], [2], [3]. The utilization of hyperspectral (HS) images is prevalent in land cover recognition and classification due to their abundant spectral and spatial information [4], [5], [6]. However, the anti-interference capability of HS images is weak, and there is also a phenomenon known as "same spectrum of foreign object" in HS image classification. These shortcomings will limit the accuracy of classification. For instance, HS images can be difficult to accurately distinguish concrete floors from buildings or a vegetation-covered roof from a flat lawn. Conversely, light detection and ranging (LiDAR) images offer robust anti-interference capability while providing precise information on the shape, size, and height of ground objects. Nonetheless, LiDAR imaging falls short when it comes to discerning objects and materials with comparable heights and sizes. The accuracy of land cover classification can be greatly improved by synergistically leveraging the respective strengths of HS and LiDAR images. Therefore, the joint classification of HS and LiDAR images has emerged as a prominent research direction in recent years.

According to the varying fusion levels of multisource remote sensing images, the HS and LiDAR classification methods can be categorized into pixel-level fusion, feature-level fusion, and decision-level fusion. Pixel-level fusion is the most fundamental form, which directly merges each pixel in the original remote sensing data. In [7], the HS and LiDAR images underwent pixel-level fusion using wavelet transform, intensity–hue–saturation (IHS) transform, and linear pixel fusion techniques before undergoing classification to minimize the data loss. However, due to significant modal disparities among heterogeneous remote sensing data, pixel-level fusion is prone to introducing noise, which limits the classification performance of models. The classification method based on feature-level fusion is widely regarded as the most prevalent technology in the field. Feature-level fusion refers to extracting features from the original remote sensing images and then integrating these extracted features to generate more precise and detailed feature information. The study conducted by Pedergnana et al. [8] utilized morphological extended attribute profiles (EAPs) to extract discriminative features from various remote sensing images, followed by classification through feature superimposition. Similarly, Hong et al. [9] employed an invariant attribute profile (IAP) to extract spatially invariant features and subsequently aggregate these features for classification purposes. The act of merely stacking features, however, is susceptible to the problem of dimensionality disaster. The aforementioned issues can be efficiently resolved through subspace learning [10], [11], [12]. For example, Rasti et al. [11] employed sparse and low-rank techniques to effectively integrate the pertinent features of HS and LiDAR

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data for accurate classification. Hong et al. [12] performed the classification by integrating the shared and specific features among different modes extracted from the manifold subspace. In contrast to pixel-level fusion and feature-level fusion, decision-level fusion represents the highest level of fusion methods. The utilization of support vector machine (SVM) with nonlinear kernel technique has gained extensive popularity [13]. In [14], data from different sources were sent to SVM after feature extraction, and then, fusion classification was carried out using naive Bayes. In [15], a differential adaptive method approach was proposed to weigh the outputs of the maximum likelihood classifier, SVM, and multinomial logistic regression classifiers to obtain the final classification result. In addition, Xia et al. [16] proposed an integrated classifier based on random forest that uses voting to obtain the final classification results. In summary, the aforementioned fusion classification methods integrate multisource data at different levels to enhance information representation and improve classification accuracy. Nevertheless, these approaches often necessitate substantial human effort and time investment in feature extraction and parameter selection processes.

The utilization of deep learning in remote sensing image processing has gained significant traction in recent years owing to its exceptional capabilities for extracting and representing features [17]. Among various deep learning techniques, the convolutional neural network (CNN) has emerged as a prominent approach in remote sensing image classification due to its inherent local perception and parameter sharing capabilities. In order to effectively integrate the advantages of multisource remote sensing images, numerous researchers have endeavored to employ CNN for the cooperative classification of such imagery [18]. Xu et al. [19] proposed a two-branch CNN for extracting spatial and spectral features from HS images, which are then fused with LiDAR branch features to obtain the final classification result. Similarly, Hang et al. [20] proposed a coupled CNN for the fusion of HS and LiDAR branches through weight sharing, followed by the utilization of a weighted summation strategy to derive the final classification outcomes.

In order to optimize the utilization of complementary information from HS and LiDAR data, numerous interaction fusion methods have also been proposed. For instance, Fang et al. [21] proposed a cross-modal enhancement module to facilitate interaction between the HS and LiDAR branches. Specifically, the HS features are utilized to augment the spectral representation of the LiDAR branch, while the LiDAR features are employed to enhance the spatial representation of the HS branch. Subsequently, these enhanced features are fused together to yield the final classification outcome. Similarly, Dong et al. [22] employed self-attention and cross-attention guidance modules to acquire more refined spectral, spatial, and elevation features. Zhang et al. [23] integrated the shared complementary features of HS and LiDAR data through the intermediate state branch and dynamically performed decision fusion on the three branches, resulting in favorable classification outcomes. Subsequently, Dong et al. [24] proposed an interpretable network that leverages a dictionary distribution to effectively align complementary spectral, spatial, and elevation information while incorporating contextual information for achieving efficient HS and LiDAR classification. Recently, Yang et al. [25] have proposed incorporating class text information into supervised learning and employed multidimensional fusion contrastive learning to establish associations between textual and visual features, thereby achieving significant improvements in classification accuracy.

Although some of the aforementioned methods have achieved satisfactory classification performance, they still inevitably have certain issues that need to be addressed. It is worth noting that both HS and LiDAR images contain rich spatial information. However, most of the existing methods ignore the common spatial features between HS and LiDAR images and their contribution to spatial information extraction, which makes it difficult to fully integrate spatial information. Building on this inspiration, the first research question (RQ) is presented, i.e., RQ1: how to effectively integrate the common spatial information from HS and LiDAR images? Furthermore, the majority of existing methods classify the fusion features directly. However, they ignore the fact that different features have significant differences in sensitivity to the same category, and it is challenging to improve the recognition ability of the model for all categories directly through simple interactive superposition of features. Consequently, this raises the second research question, RQ2: how to leverage the aware ability of different features for specific categories to achieve targeted information aggregation?

In order to address the aforementioned challenges, this article proposes an uncertainty category-aware fusion network (UCAFNet). Specifically, for RQ1, this article introduces a spatial information purification module (SIPM) in the spatial dimension, which utilizes frequency-domain information to extract and enhance the shared spatial information of HS and LiDAR images. In addition, an adaptive weighted fusion module (AWFM) is proposed to effectively integrate the spatial information of HS and LiDAR. For RQ2, we meticulously devise an uncertain category-aware fusion strategy (UCAFS). UCAFS dynamically assigns weights to the spectral, spatial, and elevation branches based on their discriminative power in identifying specific categories, thereby achieving complementary advantages. The main contributions of this study are outlined as follows.

1) A novel UCAFNet is proposed for HS and LiDAR joint classification. UCAFNet can effectively extract spectral, spatial, and elevation features based on the inherent properties of HS and LiDAR data and achieve targeted information aggregation by an information fusion strategy.
2) The proposed SIPM component utilizes frequency-domain information to extract the shared spatial information of HS and LiDAR data and dynamically adjusts their contribution through the constructed AWFM for improved integration of spatial information.
3) The proposed UCAFS performs targeted information fusion according to the ability of spectral, spatial, and elevation branches to identify specific categories and

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realizes mutual learning and complementary advantages between different features. Moreover, the proposed UCAFS can be extended to a more general multimodal joint classification.

The remaining sections of this article are structured as follows. Section II provides a comprehensive introduction to the proposed UCAFNet framework and provides the detailed explanations of the training and testing strategies. In addition, Section III describes the experimental dataset, as well as the corresponding experimental results and analysis. Finally, Section IV concludes this article.

II. Proposed Methodological Framework

In this section, we propose a UCAFNet for joint classification of HS and LiDAR images by exploiting the property that different features have different sensitivities to the same ground object. The overall framework of the proposed network is illustrated in Fig. 1, with the main body of the network being divided into two parts: feature extraction and information fusion. The feature extraction component comprises the spectral, spatial, and elevation branches collectively. Specifically, in the spectral branch, the ability of feature extraction of different band pairs is sought to obtain better spectral feature representation. In the spatial branch, the frequency-domain information is used to find the shared spatial information between HS and LiDAR, and their contributions are dynamically adjusted to obtain more refined spatial features. In addition, in the elevation branch, edge texture and vertical structure are injected to deepen the elevation information representation. Finally, in the information fusion stage, different weights were dynamically assigned to the classification confidence of each feature according to its ability to identify each category, so as to achieve targeted information aggregation.

A. Cross-Scale Spectral Feature Extraction

The HS images consist of numerous bands, each with varying sensitivity and feature extraction capabilities for different objects. To better represent the spectral information, we propose a multiscale channel attention module (MSCAM) on the spectral branch.

The input to the spectral branch consists of a given pixel sequence $X_i \in R^{1\times C}$ and a cube $P_H \in R^{m\times n\times C}$ of pixels centered at X_i in the HS dataset, where m, n, and C represent the length, width, and number of bands of the patch, respectively. When the spectral patch is used as input, it will be difficult to avoid the existence of different categories of pixels, which will affect the classifier's judgment. Therefore, before inputting the MSCAM, we perform micro-correction on the patch in order to reduce the influence of different categories of pixels on finding the relationship between spectral bands of the same category of the central pixel. The micro-corrected patch P_{mc} can be expressed as follows:

$$P_{mc} = P_H \otimes \mathrm{SA}(P_c) \tag{1}$$

where $\otimes$ multiplication represents the element multiplication, and $\mathrm{SA}(\cdot)$ represents the self-attention mechanism, which first multiplies the input W and its transpose W^T and then performs the softmax operation to obtain the normalized self-attention mask. After acquiring the micro-rectified patch, we input it into MSCAM, as illustrated in Fig. 2.

We developed MSCAM to mitigate the impact of subtle spectral differences among pixels within the same class on classification accuracy and to identify the spectral sensitivity of different classes. Specifically, the 2-D global pooling of the input P_{mc} along the spectral channel is performed, that is, we perform pixel averaging operation on each patch layer to obtain F_H with the same size as X_i, the procedure can be expressed as

$$F_H = \frac{1}{MN}\sum_{j=1}^{M}\sum_{k=1}^{N} X_{jk} \tag{2}$$

where X_{jk} represents the pixel in row j and column k of P_H. After obtaining the averaged features, 1-D convolutions with different kernel sizes (3, 5, 7) are used to establish the relationship between spectra at different distances. Subsequently, the output is passed through batch normalization (BN) and rectified linear unit (ReLU) layers to obtain different spectrally correlated features F_K, where BN and ReLU layers are used for regularization and nonlinear learning. This process can be expressed as follows:

$$F_K = \mathrm{ReLU}(\mathrm{BN}(\mathrm{Conv1}D_K(F_H))), \quad K = (3, 5, 7) \tag{3}$$

where $\mathrm{Conv1}D_K$ represents the 1-D convolution operation with kernel size K. Since the original mean feature usually has a guiding effect on classification, we fuse it with the spectral correlation feature, and the fusion process can be expressed as follows:

$$F_f = F_H + \sum F_K \tag{4}$$

where F_f represents the fused spectral features, F_H denotes the spectral characteristics obtained through average pooling, and F_K refers to the spectrally correlated features derived from F_H after applying 1-D convolution operation with convolutional kernel of size K. After that, the multiscale spectral attention mask is obtained by applying the softmax operation to the fused spectral features, and the mask is used to enhance the spectral-related information. The specific process can be described as follows:

$$F_s = X_i + X_i \otimes \mathrm{soft\,max}\left(F_f\right) \tag{5}$$

where F_s represents the enhanced spectral features and $\mathrm{soft\,max}(\cdot)$ represents the softmax operation. Finally, the final output feature F_{spe} is extracted by a 1-D convolution and a fully connected layer

$$F_{\mathrm{spe}} = \mathrm{FC}(\mathrm{Conv1}D(F_s)) \tag{6}$$

where $\mathrm{FC}(\cdot)$ represents the fully connected operation along the band direction and $\mathrm{Conv1}D(\cdot)$ represents the 1-D convolutions operation.

B. Common Spatial Feature Extraction

Both HS and LiDAR images contain valuable spatial information, but their significant modal differences make it

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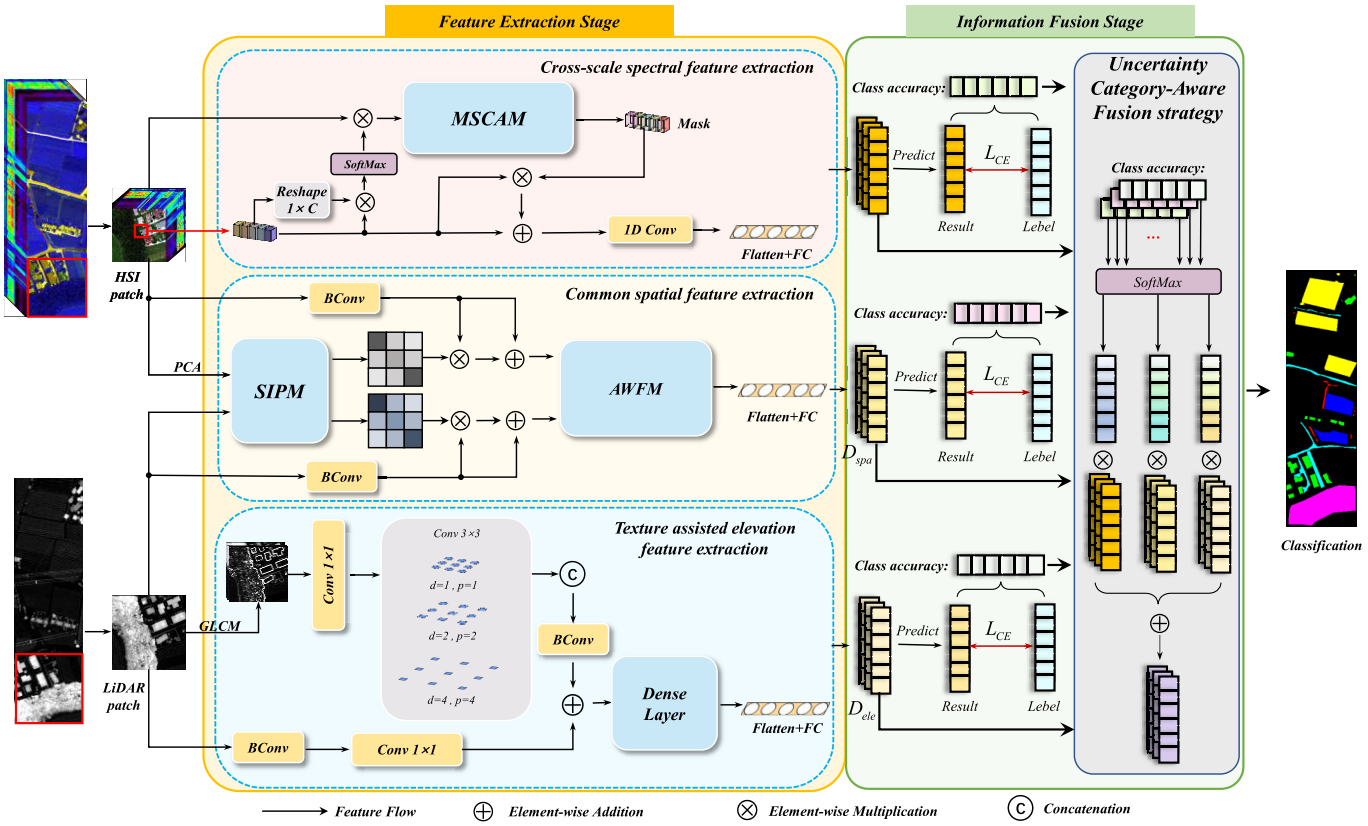


Fig. 1. Illustration of the proposed UCAFNet framework, including a feature extraction stage and an information fusion stage. Among them, the feature extraction stage consists of three branches: spectral, spatial, and elevation.

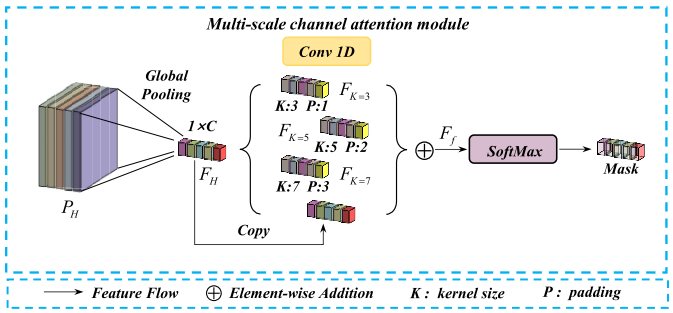


Fig. 2. Structure of MSCAM.

challenging to effectively integrate this information. To address this issue, we have developed an SIPM for the spatial branch, which can match the distinct modalities' spatial structures to guide the effective aggregation of spatial information. Furthermore, we meticulously devise an AWFM to enhance the integration of spatial features and achieve a more refined representation of spatial information. The detailed descriptions of these modules are provided as follows.

1) Spatial Information Purification Module: The SIPM, as illustrated in Fig. 3, consists of two essential components: the hash similarity encoder (HSE) and the frequency-domain spatial attention module (FDSAM). The process of purifying spatial information is elaborated upon in detail next.

The hash algorithms are widely employed for calculating image similarity due to their ability to generate unique "fingerprints" based on the image itself, which can be compared to determine the similarity between images [26]. Among various hash algorithms, the perceptual hash (pHash) algorithm [27] utilizes frequency-domain information to create distinctive fingerprints that effectively emphasize spatial structure details. Hence, we employ perceptual hashing in order to match the spatial information of HS and LiDAR data.

First, principal component analysis (PCA) is utilized to reduce the HS and LiDAR patches to a single dimension. Subsequently, the discrete cosine transform is applied to convert them into the frequency domain, yielding the frequency-domain patches D_H and D_L. They are then inputted into the HSE for obtaining the frequency-domain similarity mask, as depicted in Fig. 3(b). Initially, D_H and D_L undergo pHash encoding using the following method:

$$p_{\text{avg}} = \frac{1}{MN}\sum_{j=1}^{M}\sum_{k=1}^{N} p_{jk} \tag{7}$$

$$\hat{p}(j,k) = \begin{cases} 1, & p_{jk} \le p_{\text{avg}} \\ 0, & p_{jk} < p_{\text{avg}} \end{cases} \tag{8}$$

where M and N represent the length and the width of the frequency-domain patch, respectively; p_{jk} represents the pixel row j and column k of the patch; p_{avg} represents the mean of all pixels in the patch; and $\hat{p}(j,k)$ is the corresponding pHash encoding of p_{jk}. Next, the two groups of codes are encoded by similarity to obtain the similarity mask in the frequency domain. The similarity coding rules are given as follows:

$$\tilde{p}(j,k) = \begin{cases} 1, & \hat{p}_H(j,k) = \hat{p}_L(j,k) \\ 0, & \hat{p}_H(j,k) \neq \hat{p}_L(j,k) \end{cases} \tag{9}$$

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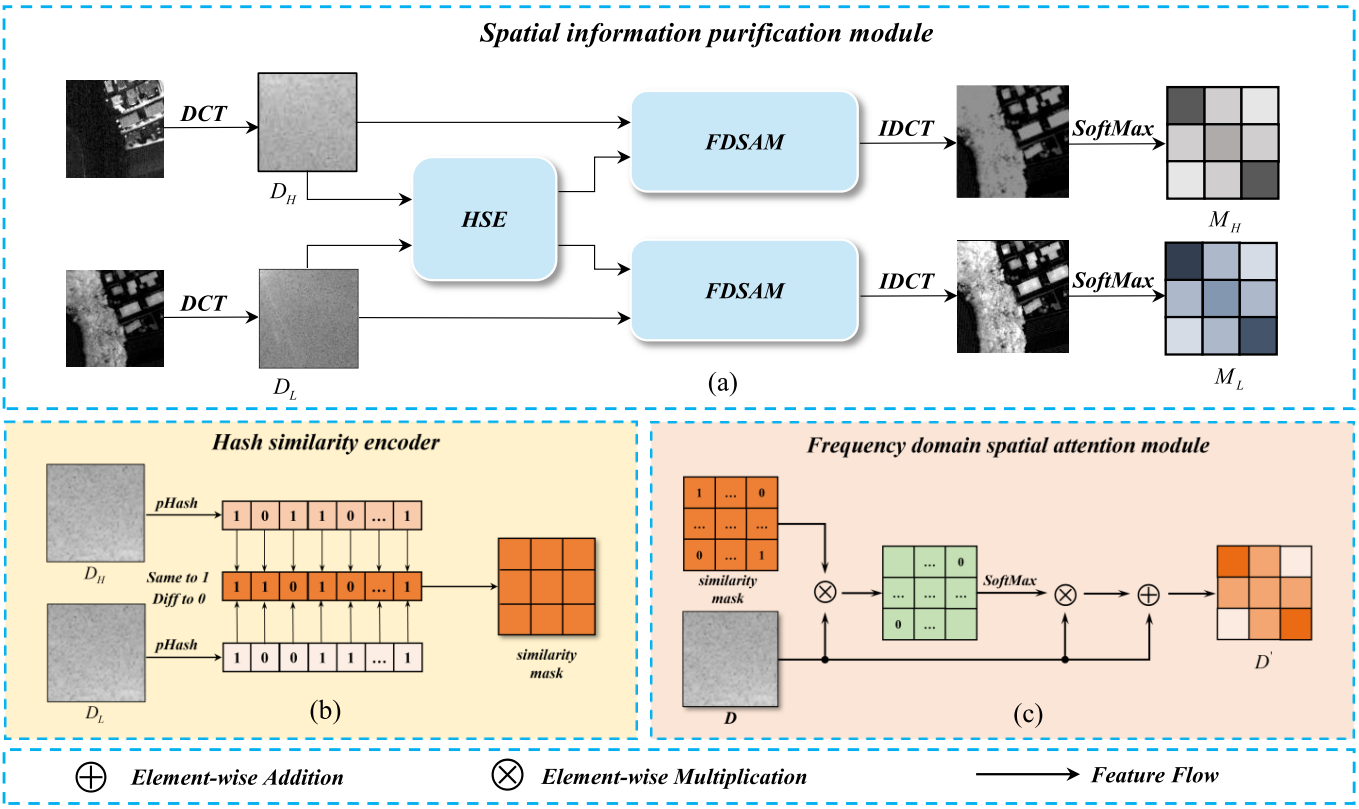


Fig. 3. Structure of the SIPM. (a) Overall framework of SIPM, which is mainly composed of two important components, HSE and FDSAM. (b) Structure of HSE. (c) Structure of FDSAM.

where the pixel in row j and column k of the HS frequency-domain patch is represented by $\hat{p}_H(j,k)$, while the corresponding LiDAR frequency-domain pixel is represented by $\hat{p}_L(j,k)$. The frequency-domain similarity code at the corresponding position is denoted as $\tilde{p}(j,k)$.

Through this encoding, we can obtain the spatially similar regions of HS and LiDAR images in the frequency domain. We emphasize the shared spatial structure information by enhancing these corresponding regions in the frequency domain, thereby providing guidance for integrating spatial information. To achieve this goal, we propose a FDSAM, as illustrated in Fig. 3(c). Specifically, the frequency-domain attention matrix is obtained by multiplying the frequency-domain similarity mask M_s with the frequency-domain patch and then normalizing it using the softmax operation. By multiplying this normalized matrix with the frequency-domain patch, enhanced spatial structure information can be acquired. Finally, to avoid information loss, the original frequency-domain patch information is injected. This process can be expressed as follows:

$$D_{H'} = D_H + D_H \otimes \text{soft}\max(D_H \otimes M_s) \tag{10}$$

$$D_{L'} = D_L + D_L \otimes \text{soft}\max(D_L \otimes M_s) \tag{11}$$

where variables $D_{H'}$ and $D_{L'}$ denote the frequency-domain patches of HS and LiDAR data, respectively, following enhancement. Then, inverse discrete cosine transform (IDCT) is performed on $D_{H'}$ and $D_{L'}$ to obtain the gray patches enhanced by shared spatial information, and the shared spatial information masks M_H and M_L are obtained after the softmax operation. These two weight masks can emphasize the shared spatial structural features of HS and LiDAR and provide better guidance for the subsequent integration of their information.

The SIPM, serving as a component of pre-aided enhancement, effectively amplifies the spatial shared features between HS and LiDAR branches. The inputs P_H and P_L of the spatial branch are fed into the convolutional layer for initial acquisition of spatial features, followed by an augmentation in the spatial structure information of both branches through shared spatial information masks. This process can be expressed as follows:

$$F_{\text{SH}} = B\text{Conv}(P_H) + B\text{Conv}(P_H) \otimes M_H \tag{12}$$

$$F_{\text{SL}} = B\text{Conv}(P_L) + B\text{Conv}(P_L) \otimes M_L \tag{13}$$

where F_{SH} and F_{SL} represent the spatial features of HS and LiDAR branches enhanced by shared spatial information, respectively; and $B\text{Conv}(\cdot)$ includes the 3×3 2-D convolution operation, BN layer, and ReLU activation function.

2) Adaptive Weighted Fusion Module: The traditional spatial feature fusion methods often overlook the significant contribution of strongly correlated features from different sources to spatial information by simply interacting or overlapping features. Moreover, the simple superposition of features also leads to redundant information. To address these limitations and achieve more effective integration of spatial information, we propose an adaptive weight fusion module (AWFM) that accurately reflects the contributions of different source features. As shown in Fig. 4, the HS stream feature F_{SH} and LiDAR stream feature F_{SL}, after undergoing shared space

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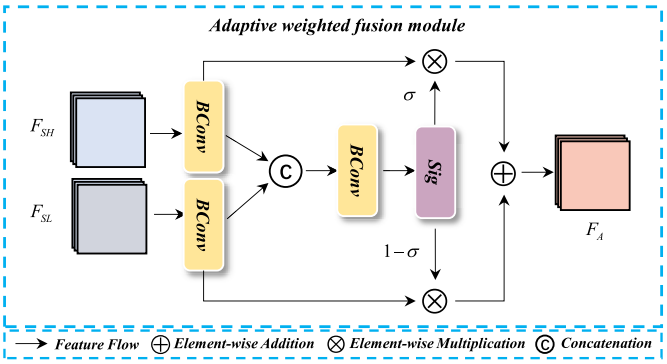


Fig. 4. Structure of AWFM.

enhancement, are inputted into the convolutional layer with shared weights to acquire profound spatial features. The subsequent outputs are then combined and inputted into a convolutional layer with a 3 × 3 kernel size. The process can be described as follows:

$$Q = B\text{Conv}(\text{Concat}(B\text{Conv}(F_{\text{SH}}), B\text{Conv}(F_{\text{SL}}))) \tag{14}$$

where Q is the fused spatial feature and Concat(·) represents the feature join operation. The sigmoid function is subsequently applied to derive the weight matrix. The spatial features of different flows are then multiplied by distinct weight matrices and fused together, resulting in the acquisition of adaptive weighted fusion spatial feature F_A, which can be mathematically expressed as follows:

$$F_A = \sigma \otimes B\text{Conv}(F_{\text{SH}}) \oplus (1-\sigma) \otimes B\text{Conv}(F_{\text{SL}}) \tag{15}$$

$$\sigma = \frac{\exp(Q)}{\exp(Q)+1} \tag{16}$$

where σ represents the weight matrix and $\oplus$ represents the element addition. The final spatial output F_{spa} is obtained through a fully connected layer. The weight matrix σ allows for adaptive adjustment of the contribution from different sources to spatial information, facilitating enhanced integration of spatial information.

C. Texture-Assisted Elevation Feature Extraction

The 3-D spatial structure and elevation information of LiDAR images can effectively distinguish the same material but different types of ground objects, so the extraction of spatial structure and elevation information is very important for research. Since the vertical structure of the ground object is closely related to the edge texture, we introduce the gray level co-occurrence matrix (GLCM) [28] in the elevation branch to help extract the edge texture and vertical structure features and combine the elevation information and edge structure texture information to improve the classification accuracy of the model.

The GLCM technique utilizes the probability distribution of gray levels among pixels in an image to extract texture features. Table I presents a comprehensive list of eight GLCM features that are effectively modeled by deep learning: mean, contrast, correlation, dissimilarity, second moment, entropy, homogeneity, and variance. In this study, we compute the

TABLE I
TEXTURE FACTORS OF GLCM

Texture feature	Formula
Mean	$Mea = \sum_{i,j=0}^{N-1} i \cdot P_{i,j}$
Contrast	$Con = \sum_{i,j=0}^{N-1} (i-j)^2 \cdot P_{i,j}$
Correlation	$Cor = \sum_{i,j=0}^{N-1} i \cdot P_{i,j} \frac{(i-Mea)(j-Mea)}{\sqrt{Var_1 \cdot Var_2}}$
Dissimilarity	$Dis = \sum_{i,j=0}^{N-1} i \cdot P_{i,j} \lvert i-j \rvert$
Second moment	$\text{Sec} = \sum_{i,j=0}^{N-1} i \cdot P_{i,j}^{\ 2}$
Entropy	$Ent = \sum_{i,j=0}^{N-1} i \cdot P_{i,j} \ln P_{i,j}$
Homogeneity	$Hom = \sum_{i,j=0}^{N-1} \frac{i \cdot P_{i,j}}{1+(i-j)^2}$
Variance	$Var = \sum_{i,j=0}^{N-1} i \cdot P_{i,j} (i-Mea)^2$

aforementioned eight GLCM texture features for LiDAR images to facilitate the extraction of vertical structure and edge texture information. To achieve this goal, we merge the statistical features of the corresponding GCLM into the elevation branch and activate neurons in each layer through 1 × 1 convolution. After that, BN and ReLU activation function were used to ensure the model stability. Meanwhile, we employ convolution operations with varying expansion and dilation rates to effectively capture the boundary information of diverse receptive fields, which can be mathematically represented as follows:

$$F_i = B\text{Conv}_{(k,d,p)}(F_{\text{GLCM}}), i \in (1,2,3) \tag{17}$$

where F_{GLCM} represents the merged GLCM features after 1 × 1 convolution, BN layer, and activation function; and $B\text{Conv}_{(k,d,p)}(\cdot)$ consists of convolution operations with kernel size k, dilation rate d, and fill rate p, along with BN layer and activation function layer; i represents the number of convolution layers. The k value is unified to 3 × 3, while d and p are set as {1, 2, 4} when i takes the values of {1, 2, 3}. For instance, in the case where $i = 1$, we set $d = 1$ and $p = 1$. This allows us to refine the boundary information to obtain the features of different receptive fields. Subsequently, concatenation and 2-D convolution operations are performed on these features to acquire more comprehensive information regarding the edges and vertical structures of ground objects. The overall process can be expressed as follows:

$$F_B = B\text{Conv}(\text{Concat}(F_1, F_2, F_3)). \tag{18}$$

To extract the elevation information from LiDAR images, we employ a 3 × 3 convolutional layer and a subsequent 1 × 1 convolutional layer on the LiDAR patch to acquire the shallow elevation feature F_{SE}. Subsequently, the shallow depth

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feature F_{SE} and edge texture feature F_B of equivalent dimensions will be integrated into a densely connected layer [29], thereby augmenting feature reusability while fully exploring edge and elevation information. The specific procedure can be expressed as follows:

$$F_E = \mathrm{Dense}(F_{\mathrm{SE}} \oplus F_B) \tag{19}$$

where $\mathrm{Dense}(\cdot)$ refers to a four-layer dense connection layer, and it can effectively preserve the elevation and texture information from each layer, thereby facilitating enhanced discrimination between elevation and texture. Ultimately, the final output F_{ele} of the elevation branch is obtained through utilization of a fully connected layer

$$F_{\mathrm{ele}} = \mathrm{FC}(F_E) \tag{20}$$

where $\mathrm{FC}(\cdot)$ represents the fully connected operation.

D. Uncertain Category-Aware Fusion Strategy

The majority of existing multimodal joint classification methods employ simple feature superposition or interaction to enhance the utilization of information from different source data. However, they overlook the fact that distinct features exhibit varying sensitivities toward the same ground object category. In other words, spectral, spatial, and elevation features contribute differently to the classifier's recognition ability for a given land cover category. Merely employing the interaction or superposition of features may undermine the advantage offered by strong features, leading to suboptimal classification results for this particular land cover category. In this article, we present the UCAFS to effectively merge the features extracted from the three branches. Specifically, UCAFS dynamically assigns different weights to the classification confidence of each feature based on their recognition ability for each category, enabling targeted information aggregation. Simultaneously, it maximizes the advantages of each branch to achieve strength complementation. Fig. 1 illustrates a schematic representation of UCAFS, and the detailed explanations of this strategy are provided next.

The UCAFS employs a two-stage fusion approach. In the initial stage, the three branches are pre-classified in the initial stage, and the weight matrix is obtained through calculation. First, the spectral, spatial, and elevation branches' features are separately inputted into three classifiers. Each classification network is a probabilistic prediction model that transforms the input into a predictive probability distribution through softmax operation. These classifiers can be trained using the maximum likelihood estimation framework, with the cross-entropy loss function being preferred

$$\mathrm{loss} = -\frac{1}{N}\sum_{i=1}^{N} Y^i \log\left(p^i\right) \tag{21}$$

where N represents the number of training samples, Y^i denotes the ith class in the class label, and p^i is the probability of predicting the ith class after softmax operation. The total network loss L_{total} is composed of the losses $\mathrm{loss}_{\mathrm{spe}}$, $\mathrm{loss}_{\mathrm{spa}}$, and $\mathrm{loss}_{\mathrm{ele}}$ of the spectral, spatial, and elevation branches together

$$L_{\mathrm{total}} = \lambda_1 \mathrm{loss}_{\mathrm{spe}} + \lambda_2 \mathrm{loss}_{\mathrm{spa}} + \lambda_3 \mathrm{loss}_{\mathrm{ele}} \tag{22}$$

where λ_1, λ_2, and λ_3 represent the weight parameters of $\mathrm{loss}_{\mathrm{spe}}$, $\mathrm{loss}_{\mathrm{spa}}$, and $\mathrm{loss}_{\mathrm{ele}}$ respectively, and we empirically set all of them to 1 for satisfactory classification effect. The classifier of the three branches can be utilized to obtain prediction results, and by comparing them with the labels, the average accuracy for each branch can be calculated. The classification accuracy for each category within each branch is determined by comparing the predicted results obtained from the three classifiers with the corresponding labels

$$\begin{cases} \mathrm{acc}_i^t = \mathrm{pred}_i^{\mathrm{true}}/gt_i, i = 1, \ldots, N \\ \mathrm{ACC}_{\mathrm{spe}} = \left[\mathrm{acc}_i^{\mathrm{spe}}, \ldots, \mathrm{acc}_N^{\mathrm{spe}}\right] \\ \mathrm{ACC}_{\mathrm{spa}} = \left[\mathrm{acc}_i^{\mathrm{spa}}, \ldots, \mathrm{acc}_N^{\mathrm{spa}}\right] \\ \mathrm{ACC}_{\mathrm{ele}} = \left[\mathrm{acc}_i^{\mathrm{ele}}, \ldots, \mathrm{acc}_N^{\mathrm{ele}}\right] \end{cases} \tag{23}$$

where acc_i^m represents the classification accuracy of class i of the t branch, $\mathrm{pred}_i^{\mathrm{true}}$ represents the number of correct classification of class i, and gt_i represents the number of samples of class i. $\mathrm{ACC}_{\mathrm{spe}}$, $\mathrm{ACC}_{\mathrm{spa}}$, and $\mathrm{ACC}_{\mathrm{ele}}$ denote the class accuracy matrices for spectral, spatial, and elevation branches, respectively. Then, the accuracy of each class corresponding to the three branches is normalized, the weight sum of each class of the three branches is equal to 1, and three corresponding weight matrices are obtained (24), as shown at the bottom of the next page, where ∂_{spe}, ∂_{spa}, and ∂_{ele} denote the discriminative information weight matrices of spectral, spatial, and elevation branches, respectively, and $\mathrm{ACC}_{\mathrm{spe}}^i$ denotes the ith vector in $\mathrm{ACC}_{\mathrm{spe}}$. The discriminative information weight matrices can effectively capture the difference of discriminative ability of different features for each category, so as to provide more precise guidance for targeted information aggregation.

Subsequently, in the second stage, we perform weighted fusion on the output features F_{spe}, F_{spa}, and F_{ele} of the three branches to obtain comprehensive features F_{out} with targeted information, and the final classification result is obtained through it

$$F_{\mathrm{out}} = \partial_{\mathrm{spe}} \cdot F_{\mathrm{spe}} + \partial_{\mathrm{spa}} \cdot F_{\mathrm{spa}} + \partial_{\mathrm{ele}} \cdot F_{\mathrm{ele}}. \tag{25}$$

III. Experiment and Analysis

In this section, we conducted experiments to validate the performance of the proposed UCAFNet. First, we provide a description of the three public remote sensing datasets utilized in our experiments. Second, we present the experimental setup. Next, we compare and analyze the classification results of UCAFNet with various classical and state-of-the-art methods. Finally, ablation experiments were conducted to demonstrate the effectiveness of the proposed module.

A. Description of Experimental Data

The present study employs three publicly available HS and corresponding LiDAR datasets, namely, the Houston dataset [30], Trento dataset [31], and Muufl dataset [32], [33],

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TABLE II
NUMBER OF TRAINING AND TESTING SAMPLES FOR HOUSTON DATASET

No.	Class	Number of train samples	Number of test samples
1	Health grass	198	1251
2	Stressed grass	190	1254
3	Synthetic grass	192	697
4	Trees	188	1244
5	Soil	186	1242
6	Water	182	325
7	Residential	196	1268
8	Commercial	191	1244
9	Road	193	1252
10	Highway	191	1227
11	Railway	181	1235
12	Parking lot 1	192	1233
13	Parking lot 2	184	469
14	Tennis court	181	428
15	Running track	187	660
	Total	2832	15029

TABLE III
NUMBER OF TRAINING AND TESTING SAMPLES FOR TRENTO DATASET

No.	Class	Number of train samples	Number of test samples
1	Apple trees	129	4034
2	Building	125	2903
3	Ground	105	479
4	Woods	154	9123
5	Vineyard	184	10501
6	Roads	122	3174
	Total	819	30214

TABLE IV
NUMBER OF TRAINING AND TESTING SAMPLES FOR MUUFL DATASET

No.	Class	Number of train samples	Number of test samples
1	Trees	150	23246
2	Mostly grass	150	4270
3	Mixed ground surface	150	6882
4	Dirt and sand	150	1826
5	Road	150	6687
6	Water	150	466
7	Building Shadow	150	2233
8	Building	150	6240
9	Sidewalk	150	1385
10	Yellow curb	150	183
11	Cloth panels	150	269
	Total	1650	53687

for experimental purposes. The detailed descriptions of these datasets are provided as follows.

1) Houston Dataset: The first dataset was acquired during the summer of 2012 within the premises of the University of Houston and its surrounding areas, as referenced in the 2013 geoscience and remote sensing society (GRSS) data fusion contest. This dataset comprises HS images with dimensions of 349 × 1905 pixels, accompanied by LiDAR data, both possessing a spatial resolution of 2.5 m. The HS image encompasses 144 spectral bands spanning wavelengths from 0.38 to 1.05 μm, while the LiDAR data consist of one band. Within this dataset, there are a total of 15 categories, and Table II provides specific details regarding sample distribution for training and testing purposes.

2) Trento Dataset: The second dataset was acquired in a rural area of southern Trento, Italy, and comprises HS images and LiDAR data with a spatial resolution of 1 m and dimensions of 166 × 600 pixels. The HS images were captured by the AISA Eagle sensor, which offers 63 spectral bands covering wavelengths ranging from 0.42 to 0.99 μm, while the LiDAR data were collected using the Optech airborne lidar terrain mapper (ALTM) 3100EA sensor that operates within a single spectral band. This dataset encompasses six distinct categories, and Table III presents the detailed information on each category along with the corresponding number of training and testing samples.

3) Muufl Dataset: The final dataset was obtained at the University of Mississippi Gulf Coast campus and comprises HS images and LiDAR data, with a spatial resolution of 1 m and a size of 325 × 220 pixels. The HS images consist of 64 spectral bands ranging from 375 to 1.05 μm, while the LiDAR data consist of two spectral bands. This dataset encompasses 11 distinct categories, and Table IV provides the details on the number of samples used for training and testing.

B. Experimental Setup

All deep learning networks were implemented using the PyTorch framework. In addition, we employed the Adam

$$\begin{cases} \partial_{\text{spe}} = \left[\dfrac{\text{ACC}^1_{\text{spe}}}{\text{ACC}^1_{\text{spe}} + \text{ACC}^1_{\text{spa}} + \text{ACC}^1_{\text{ele}}}, \ldots, \dfrac{\text{ACC}^N_{\text{spe}}}{\text{ACC}^N_{\text{spe}} + \text{ACC}^N_{\text{spa}} + \text{ACC}^N_{\text{ele}}} \right] \\ \partial_{\text{spa}} = \left[\dfrac{\text{ACC}^1_{\text{spa}}}{\text{ACC}^1_{\text{spe}} + \text{ACC}^1_{\text{spa}} + \text{ACC}^1_{\text{ele}}}, \ldots, \dfrac{\text{ACC}^N_{\text{spa}}}{\text{ACC}^N_{\text{spe}} + \text{ACC}^N_{\text{spa}} + \text{ACC}^N_{\text{ele}}} \right] \\ \partial_{\text{ele}} = \left[\dfrac{\text{ACC}^1_{\text{ele}}}{\text{ACC}^1_{\text{spe}} + \text{ACC}^1_{\text{spa}} + \text{ACC}^1_{\text{ele}}}, \ldots, \dfrac{\text{ACC}^N_{\text{ele}}}{\text{ACC}^N_{\text{spe}} + \text{ACC}^N_{\text{spa}} + \text{ACC}^N_{\text{ele}}} \right] \end{cases} \tag{24}$$

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optimization algorithm to adaptively optimize these models. The batch size was set to 64, while the learning rate and epoch were set to 0.001 and 150, respectively, for conducting the experiments.

C. Experimental Results and Analysis

To evaluate the classification performance of UCAFNet, we compare the classification results of the proposed method with some classical and recent fusion classification methods. These methods include the classical SVM [34], RF [35], S2FL [12], and CoSpace-$\ell 1$ [36] algorithms, and the recent popular deep learning methods, such as two-branch CNN [19], coupled CNN [20], MAHiDFNet [37], and FDCFNet [23]. Among them, SVM and RF are single-source input models, and in order to ensure the fairness of the experiment, we use the classical multisource data fusion method to fuse the HS and LiDAR data and input the model. Meanwhile, in order to ensure the comparability and fairness of the experiments, all experiments use the same training and testing samples and set optimal experimental parameters as reported in their respective articles.

In the experiment, overall classification accuracy (OA), average classification accuracy (AA), and Kappa coefficient were used as quantitative evaluation indexes. The OA represents the ratio of correctly classified samples to all samples, while AA represents the average classification accuracy of each class. The Kappa coefficient serves as an indicator for assessing classification consistency and can be expressed using the following formula:

$$\text{Kappa} = \frac{\text{OA} - p_e}{1 - p_e} \tag{26}$$

where

$$p_e = \frac{\sum_{i=1}^{L} a_i \times b_i}{n^2} \tag{27}$$

where a_i represents the actual number of samples in class i, b_i represents the predicted number of samples in class i, L is the total number of classes, and n is the total number of samples. The aforementioned performance indicators can be utilized for assessing the model's classification performance, with higher values denoting superior classification efficacy.

To ensure experimental validity, all results are taken from the average of five independent experiments. Figs. 5–7 show the visual classification results on the three datasets, and Tables V–VII present the quantitative classification results, where the optimal results are shown in bold. The visualization results and quantitative indicators of each comparison method on the Houston dataset are presented in Fig. 5 and Table V. Due to the large scale of Houston and the scattered distribution of labeled sample points, it is possible for the difference within the same category to increase, while the similarity between different categories also increases, leading to misclassification. In Fig. 5, it can be observed that other methods easily misclassify railway as highway, whereas our proposed UCAFNet achieves considerable classification accuracy by leveraging a UCAFS. This strategy effectively exploits discriminative information from different features, significantly enhancing class identification capability. Meanwhile, as illustrated in Table V, the AA of the proposed method reaches a considerable 96.92%, which is 5.99%, 1%, 5.56%, and 0.4% higher than the latest deep learning methods two-branch CNN, coupled CNN MAHiDFNet, and FDCFNet, respectively. The OA was 9.62%, 1.46%, 7.22%, and 0.19% higher, and the Kappa coefficient was 10.42%, 1.58%, 7.80%, and 0.20% higher.

The visual classification results and quantitative evaluation indicators of different methods on the Trento dataset are presented in Fig. 6 and Table VI. In this dataset, samples belonging to various categories are predominantly concentrated in distinct areas; however, there exist some instances where samples from different categories are closely interconnected, such as buildings and roads. Due to their homogeneity, effectively distinguishing between these two types of ground objects becomes challenging. As can be seen in the enlarged area in Fig. 6, many deep learning approaches tend to misclassify the intricate junctions between buildings and roads to a significant extent. With the assistance of SIPM and the edge texture injected by the elevation branch, UCAFNet has higher classification accuracy in these difficult regions, and the boundary of the classification is clearer. Compared with other deep learning methods, the OA of UCAFNet is 5.08%, 0.44%, 0.67%, and 0.15% higher, the AA is 9.66%, 0.94%, 1.21%, and 0.26% higher, and the Kappa coefficient is 6.69%, 0.59%, 0.89%, and 0.2% higher.

Table VII shows the classification quantitative index results of different methods on the Muufl dataset. Compared with the latest deep learning methods, i.e., the two-branch CNN, coupled CNN, MAHiDFNet, and FDCFNet, the proposed UCAFNet outperforms better with a higher OA by 4.45%, 3.92%, 1.29%, and 0.22%, respectively. In addition, the AA surpasses them by 14.4%, 2.19%, 14.34%, and 0.83%, while the Kappa coefficient exceeds them by 5.81%, 4.97%, 1.69%, and 0.32%. Fig. 7 illustrates the classification results of various methods on the Muufl dataset where it can be observed that the classification of trees and mixed ground surface in the enlarged area by the proposed method is almost all correct, and the boundary is clearer, which is very close to the ground truth, which fully proves the superior performance of the model.

D. Parameter Tuning and Ablation Experiment

The network's classification performance is closely tied to the hyperparameters and the designed network architecture. Thus, in this section, we delve into three predefined hyperparameters and demonstrate the efficacy of the five modules incorporated within the network.

1) *Analysis on Network Input Patch Size:* The size of the HS and LiDAR patch input to the network will impact the amount of information fed into the network. If the input patch size is too small, it can lead to significant information loss. Conversely, if the size of the input patch is too large, it may increase the amount of calculation and may also affect the information of the given pixel. Therefore, to investigate the impact of network input patch size on network classification performance, we systematically adjust the patch size from a candidate

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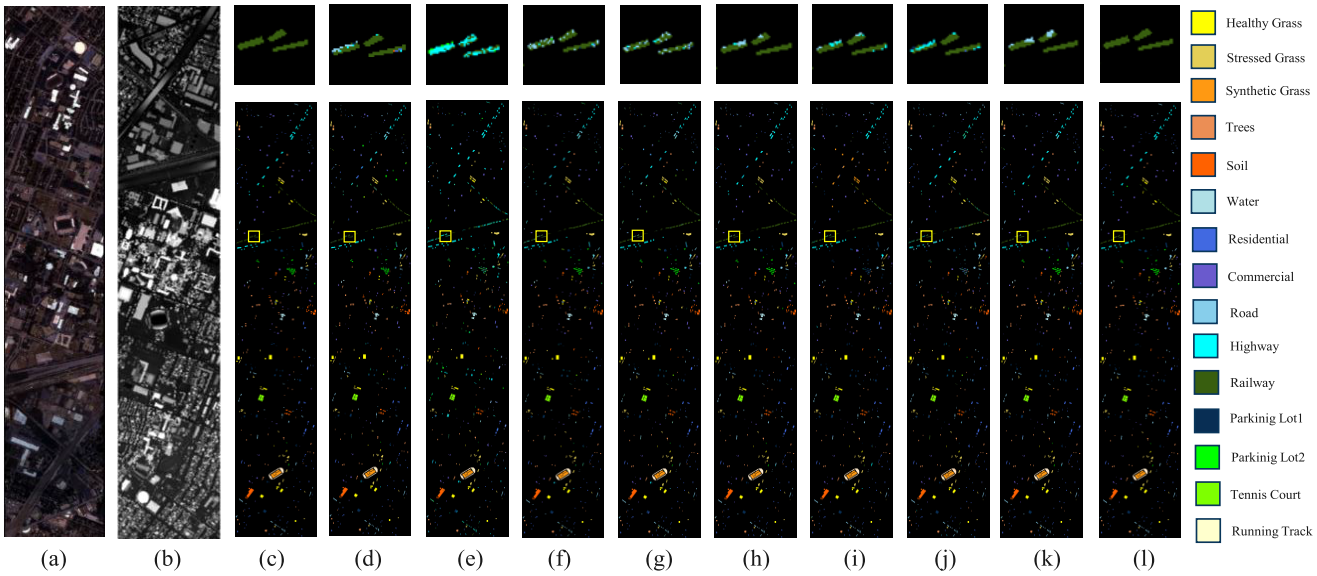


Fig. 5. Classification maps of different methods on the Houston dataset. (a) HS pseudo-color image of spectral bands of 45, 25, and 10 as R, G, and B, respectively. (b) LiDAR images. (c) Ground-truth map. (d) SVM. (e) RF. (f) S2FL. (g) CoSpace-ℓ1. (h) Two-branch CNN. (i) Coupled CNN. (j) MAHiDNet. (k) FDCFNet. (l) Proposed.

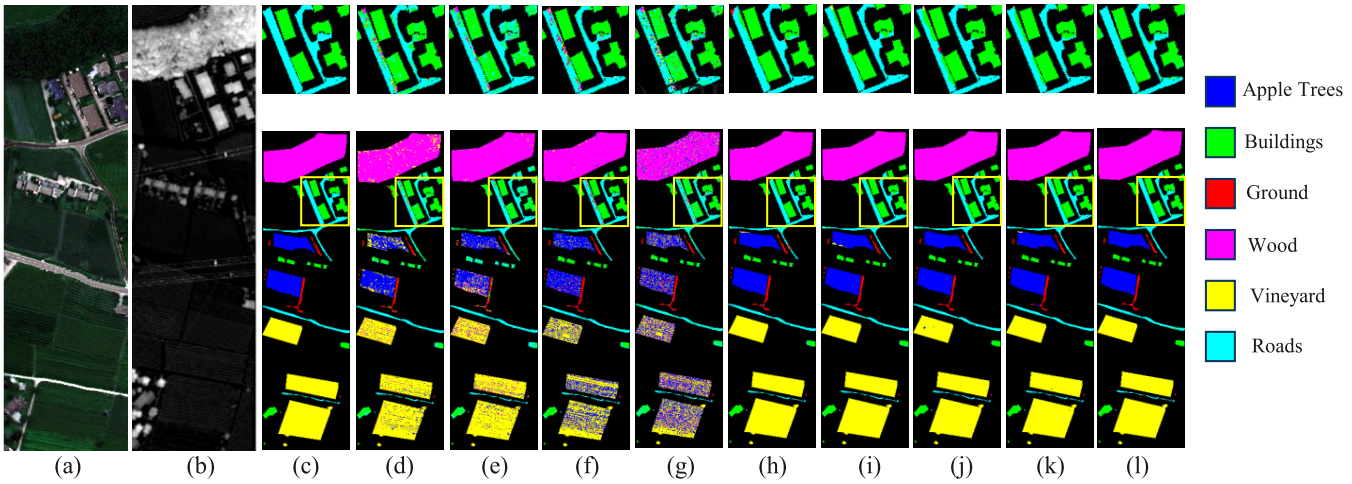


Fig. 6. Classification maps of different methods on the Trento dataset. (a) HS pseudo-color image of spectral bands of 25, 18, and 10 as R, G, and B, respectively. (b) LiDAR images. (c) Ground-truth map. (d) SVM. (e) RF. (f) S2FL. (g) CoSpace-ℓ1. (h) Two-branch CNN. (i) Coupled CNN. (j) MAHiDNet. (k) FDCFNet. (l) Proposed.

set $\{7 \times 7, 9 \times 9, 11 \times 11, 13 \times 13, 15 \times 15, 17 \times 17, 19 \times 19\}$ while keeping other parameters constant. Fig. 8(a) illustrates the OA achieved by the model with different patch sizes across three datasets. The experimental results demonstrate that optimal classification performance is attained when using an 11×11 patch size.

2) *Analysis on Network Learning Rate:* The learning rate of the network is a crucial hyperparameter in deep learning, representing the step size by which the gradient moves toward the optimal solution of the loss function during each iteration. When the learning rate is too small, convergence becomes sluggish; conversely, if it is too large, oscillations or even an increase in loss may occur. Therefore, to obtain the optimal value, we kept all other network parameters unchanged and conducted experiments on three datasets while adjusting the learning rate from 0.00001 to 1. Fig. 8(b) demonstrates that setting the learning rate to 0.001 yields satisfactory convergence of our model.

3) *Analysis on the Weight Parameters*: In (22), the loss function of the model contains three hyperparameters (i.e., λ_1–λ_3). To test their impact on the classification performance of the model, we initially fix λ_2 and λ_3 constant while varying λ_1 from a candidate set $\{0.001, 0.01, 0.1, 1\}$ to determine the optimal value for λ_1. Subsequently, with the optimal value for λ_1 fixed along with λ_3, we vary λ_2 from the same candidate set. Finally, keeping both the optimal values for λ_1 and λ_2 unchanged, we explore different values of λ_3 in the same candidate set. Fig. 9 shows the OA obtained by the proposed UCAFNet for different λ_1–λ_3 values on Houston, Trento, and Muufl datasets. The experimental results demonstrate that the model achieves

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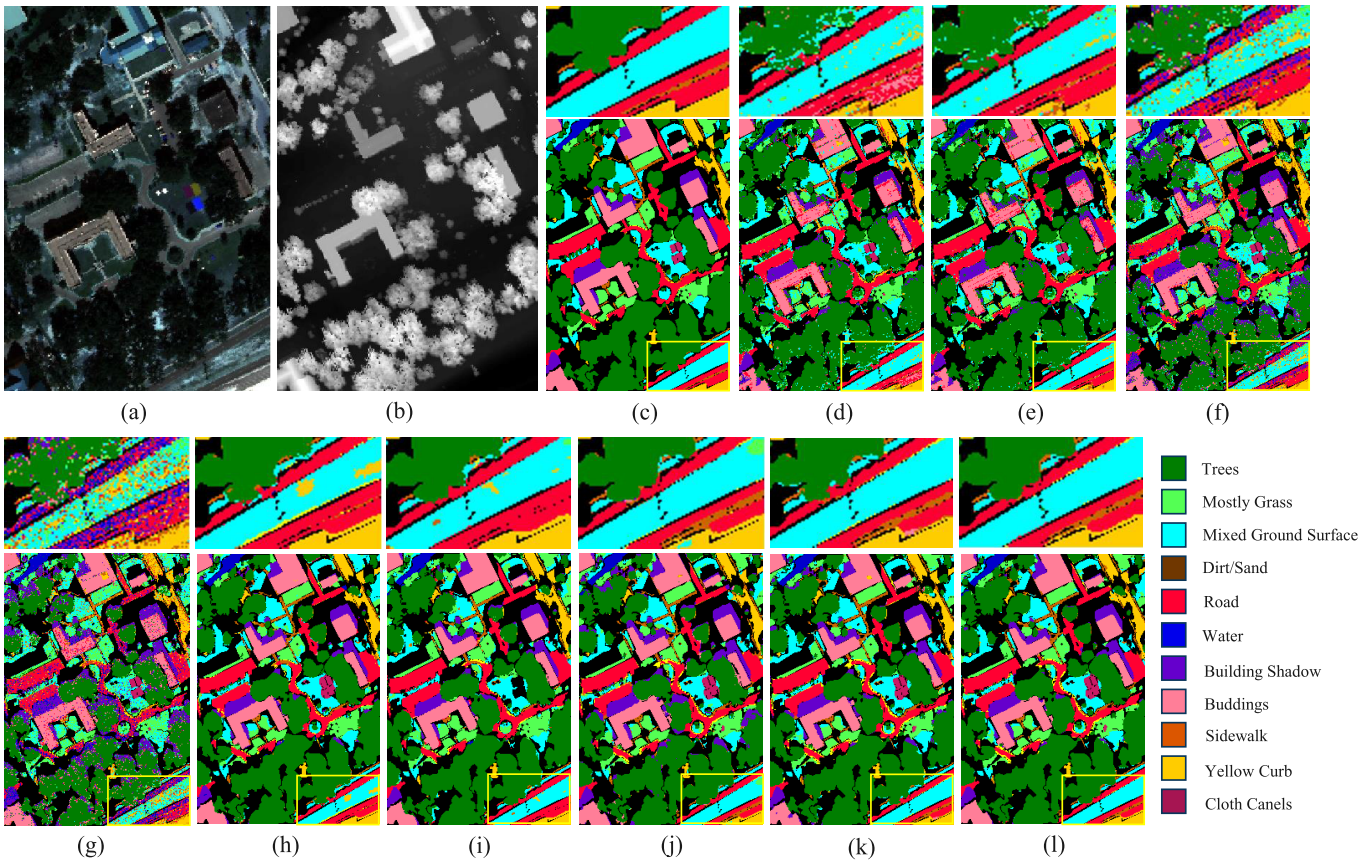


Fig. 7. Classification maps of different methods on the Muufl dataset. (a) HS pseudo-color image of spectral bands of 30, 20, and 10 as R, G, and B, respectively. (b) LiDAR images. (c) Ground-truth map. (d) SVM. (e) RF. (f) S2FL. (g) CoSpace-ℓ1. (h) Two-branch CNN. (i) Coupled CNN. (j) MAHiDNet. (k) FDCFNet. (l) Proposed.

TABLE V
CLASSIFICATION ACCURACY OF HOUSTON DATA BY DIFFERENT METHODS AND KAPPA COEFFICIENT

No.	Class	SVM	RF	S2FL	CoSpace-ℓ1	two-branch CNN	Coupled CNN	MAHiDFNet	FDCFNet	Proposed
1	Health grass	94.06	92.75	90.57	89.45	98.80	85.61	98.53	97.45	93.76
2	Stressed grass	95.07	95.39	97.69	97.05	84.22	100.00	92.87	100.00	99.44
3	Synthetic grass	100	95.12	100.00	100.00	96.27	97.85	91.11	96.84	98.42
4	Trees	98.51	99.01	98.39	98.47	95.22	99.92	98.10	99.60	99.28
5	Soil	95.77	96.02	99.36	99.36	97.56	100.00	98.38	100.00	100.00
6	Water	79.37	82.54	99.38	99.69	98.33	100.00	95.58	100.00	98.15
7	Residential	92.44	90.41	80.52	80.52	98.01	91.01	99.15	94.64	94.48
8	Commercial	80.53	68.35	68.57	89.39	86.34	93.73	80.94	97.11	96.30
9	Road	79.9	70.02	69.89	64.86	74.04	94.73	98.04	94.68	94.49
10	Highway	93.78	78.49	70.01	66.75	90.05	91.93	72.81	94.38	95.76
11	Railway	88.31	82.05	88.34	88.42	92.44	96.60	72.71	98.30	99.76
12	Parking lot 1	62.12	67.56	84.27	84.02	57.50	94.24	76.80	93.03	92.78
13	Parking lot 2	37.14	12.75	83.58	78.68	100.00	95.74	95.80	91.82	94.46
14	Tennis court	98.31	87.47	100.00	100.00	96.39	99.07	99.53	99.77	97.20
15	Running track	98.75	86.41	98.94	98.18	98.79	98.33	100.00	90.37	99.55
	OA(%)	87.61	82.50	86.81	87.52	87.18	95.34	89.58	96.61	**96.80**
	AA(%)	86.27	80.29	88.63	88.99	90.93	95.92	91.36	96.53	**96.92**
	Kappa×100	86.60	81.05	85.75	86.51	86.12	94.96	88.74	96.34	**96.54**

optimal classification performance when λ_1–λ_3 are set to 1.

4) *Results of Ablation Experiments:* The ablation experiment results of the module are shown in

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TABLE VI
CLASSIFICATION ACCURACY OF TRENTO DATA BY DIFFERENT METHODS AND KAPPA COEFFICIENT

No.	Class	SVM	RF	S2FL	CoSpace-ℓ1	two-branch CNN	Coupled CNN	MAHiDFNet	FDCFNet	Proposed
1	Apple trees	81.91	78.69	71.02	86.04	77.94	98.88	99.19	100.00	100.00
2	Building	91.82	85.01	82.57	95.83	89.29	97.83	88.92	98.29	98.52
3	Ground	93.55	94.62	92.07	95.62	70.03	97.49	97.53	99.48	99.16
4	Woods	98.07	94.86	86.69	98.73	100.00	99.89	99.98	100.00	100.00
5	Vineyard	92.11	90.21	48.90	63.56	99.60	100.00	99.90	100.00	99.80
6	Roads	80.55	77.75	77.57	89.16	97.74	92.82	99.78	93.23	95.05
	OA(%)	90.36	88.34	70.20	83.48	94.18	98.82	98.59	99.11	**99.26**
	AA(%)	88.00	86.86	76.47	88.16	89.10	97.82	97.55	98.50	**98.76**
	Kappa×100	87.13	84.40	61.91	78.66	92.32	98.42	98.12	98.81	**99.01**

TABLE VII
CLASSIFICATION ACCURACY OF MUUFL DATA BY DIFFERENT METHODS AND KAPPA COEFFICIENT

No.	Class	SVM	RF	S2FL	CoSpace-ℓ1	two-branch CNN	Coupled CNN	MAHiDFNet	FDCFNet	Proposed
1	Trees	95.15	92.83	72.44	78.55	98.34	91.67	98.29	96.46	95.27
2	Mostly grass	74.34	78.47	68.67	74.36	81.54	94.70	86.63	84.29	90.84
3	Mixed ground surface	80.84	74.57	53.66	63.18	73.04	67.63	83.66	84.82	85.76
4	Dirt and sand	85.15	77.36	71.08	81.00	85.05	87.90	83.65	95.51	93.81
5	Road	93.05	89.01	58.47	77.99	84.54	88.26	96.67	89.43	90.73
6	Water	81.19	73.89	94.42	96.57	85.98	99.79	78.84	98.93	100.00
7	Building Shadow	65.6	69.99	76.62	83.65	80.29	97.49	74.85	96.15	96.37
8	Building	84.27	82.17	80.77	86.17	98.50	96.44	98.72	96.55	96.03
9	Sidewalk	48.25	35.89	51.05	63.25	78.28	91.77	71.93	87.65	88.45
10	Yellow curb	66.29	5.62	95.63	96.72	30.72	96.72	40.40	97.27	98.91
11	Cloth panels	77.39	86.59	97.40	98.88	81.13	99.26	64.43	99.63	99.63
	OA(%)	87.05	83.95	68.93	77.28	88.67	89.20	91.83	92.90	**93.12**
	AA(%)	77.41	69.4	74.56	81.85	79.76	91.97	79.82	93.33	**94.16**
	Kappa×100	82.81	78.82	61.35	71.30	85.19	86.03	89.31	90.68	**91.00**

Tables VIII and IX, and the results are analyzed as follows.

In the proposed network, we design three modules MSCAM, SIPM, and AWFM in the feature extraction stage. To validate the efficacy of these modules individually, we conducted experiments by removing each module while keeping other framework conditions unchanged. In addition, to enhance the credibility of our experimental results, we also performed experiments where all three modules were removed simultaneously. As depicted in Table VIII, without MSCAM, SIPM, and AWFM individually on the Houston dataset resulted in a reduction of 2.11%, 1.83%, and 0.77%, respectively in OA, whereas on the Trento dataset, it led to reductions of 2.18%, 2.28%, and 1.57%. On the Muufl dataset as well, OA was reduced by 2.83%, 2.4%, and 1.93% for each module removal case mentioned above. This unequivocally demonstrates that MSCAM, SIPM, and AWFM can effectively obtain discriminative information from the feature extraction stage. In addition, we performed ablation experiments on GLCM. It can be seen that after removing GLCM, OA decreases by 0.59%, 0.5%, and 1.04% on the three datasets. Also, the corresponding AA and Kappa coefficients also decreased accordingly. This fully proves that adding texture features can improve the classification performance of the model. Moreover, the experimental results obtained when removing all four modules concurrently further substantiate their positive impact on enhancing model's classification performance.

In addition, we performed ablation experiments on UCAFS. The UCAFS aims to take advantage of the fact that different features have different sensitivities to the same ground object and aggregate strong discriminative information to improve the classification accuracy of each class. Table IX shows the classification performance of the model on the three datasets before and after the introduction of UCAFS strategy, and the model outputs the fusion features by simple superposition when the strategy is not introduced. The results show that the introduction of UCAFS strategy on the three datasets can improve the OA by 2.24%, 2.72%, and 3.01%. In particular, the AA index is significantly improved, which is increased by 2.49%, 2.75%, and 3.81%, respectively, on Houston, Trento,

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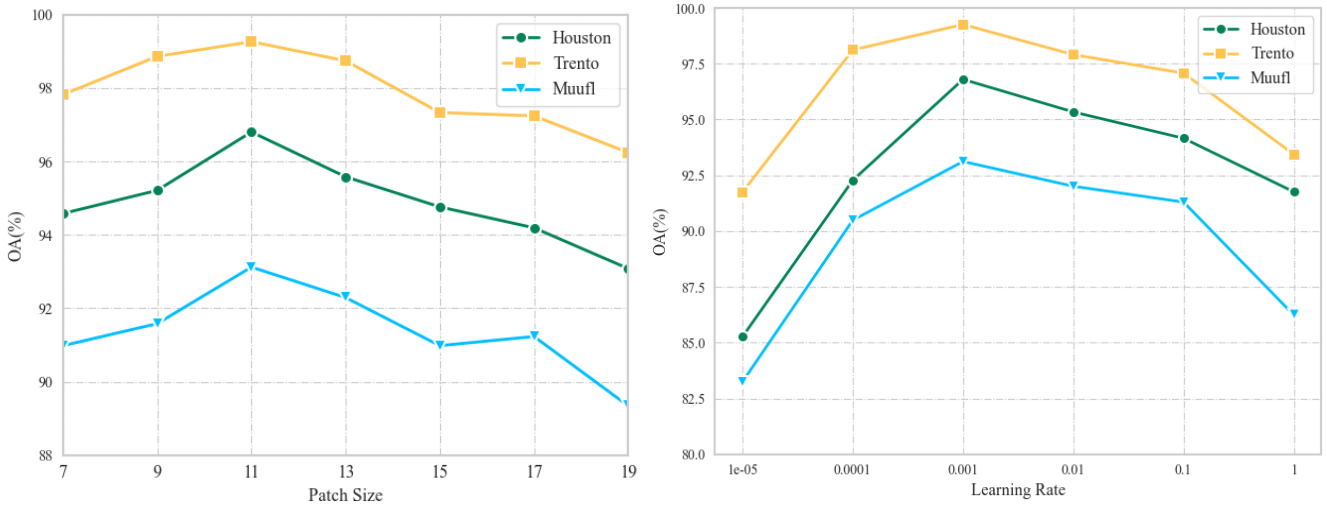


Fig. 8. Impact of different hyperparameters on the overall accuracy (OA) of three datasets was examined. (a) Effect of input patch size on model performance (OA) by UCAFNet. (b) Effect of learning rate on model performance (OA) by UCAFNet.

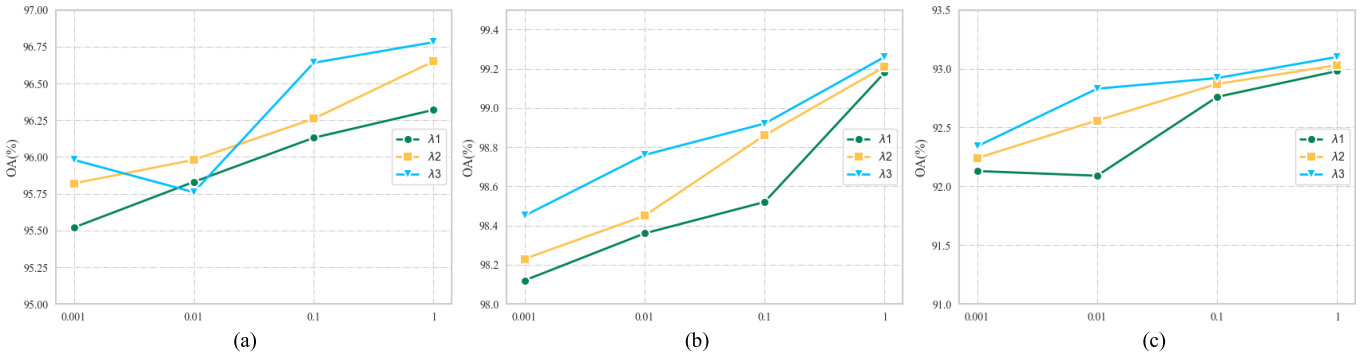


Fig. 9. Effects of weight parameters λ_1–λ_3 on the classification performance achieved by the UCAFNet on three data. (a) Houston data. (b) Trento data. (c) Muufl data.

TABLE VIII

RESULTS OF CLASSIFICATION IN ABLATION STUDIES FOR MSCAM, SIPM, AWFM, AND GLCM OF THE UCAFNET

Dataset	Methods	OA(%)	AA(%)	Kappa(%)
Houston	Base	91.27	91.48	91.02
	w/o MSCAM	94.69	94.82	95.27
	w/o SIPM	94.97	95.04	94.88
	w/o AWFM	96.03	96.25	95.76
	w/o GLCM	96.21	96.42	95.94
	UCAFNet	**96.80**	**96.92**	**96.54**
Trento	Base	94.77	94.23	94.48
	w/o MSCAM	97.08	96.65	96.84
	w/o SIPM	96.98	96.54	96.72
	w/o AWFM	97.69	97.18	97.49
	w/o GLCM	98.76	98.15	98.23
	UCAFNet	**99.26**	**98.76**	**99.01**
Muufl	Base	86.03	85.92	85.29
	w/o MSCAM	90.29	89.98	89.27
	w/o SIPM	90.72	91.07	90.28
	w/o AWFM	91.27	91.75	90.34
	w/o GLCM	92.08	92.96	90.08
	UCAFNet	**93.12**	**94.16**	**91.00**

and Muufl datasets, which fully proves the strong promotion effect of UCAFS strategy in improving the classification accuracy of each type of ground objects.

TABLE IX

RESULT OF CLASSIFICATION IN ABLATION STUDIES FOR UCAFS OF THE UCAFNET

Dataset	Methods	OA(%)	AA(%)	Kappa(%)
Houston	w/o UCAFS	94.56	94.43	94.18
	UCAFNet	**96.80**	**96.92**	**96.54**
Trento	w/o UCAFS	96.54	96.01	96.31
	UCAFNet	**99.26**	**98.76**	**99.01**
Muufl	w/o UCAFS	90.11	90.35	89.29
	UCAFNet	**93.12**	**94.16**	**91.00**

IV. CONCLUSION

The present study introduces a multisource information fusion classification network, namely, UCAFNet, for land cover classification of HS and LiDAR data. UCAFNet employs UCAFS to determine the sensitivity of different features toward specific ground object categories and utilizes these sensitivities to emphasize the significant contribution of dominant features. Moreover, an MSCAM is devised to establish the relationship between spectra in the spectral branch. In the spatial branch, SIPM is designed to capture common spatial information from HS and LiDAR in the frequency domain, while an AWFM is employed for comprehensive integration

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of spatial information from both data sources. In addition, texture information is incorporated into the elevation branch to aid in extracting elevation details for improved identification of edge objects. The experimental results demonstrate that UCAFNet surpasses other state-of-the-art methods. The future work will focus on the application of this method to other multisource remote sensing joint classifications, such as HS and multispectral image fusion classification and multispectral and panchromatic image fusion classification. This integration algorithm will be achieved by utilizing the unique recognition capabilities possessed by different sensors for various categories of ground objects.

REFERENCES

[1] Y. Huang et al., "Cross-scene wetland mapping on hyperspectral remote sensing images using adversarial domain adaptation network," *ISPRS J. Photogramm. Remote Sens.*, vol. 203, pp. 37–54, Sep. 2023.

[2] Q. Liu, K. Ren, X. Meng, and F. Shao, "Domain adaptive cross reconstruction for change detection of heterogeneous remote sensing images via a feedback guidance mechanism," *IEEE Trans. Geosci. Remote Sens.*, vol. 61, 2023, Art. no. 4507216.

[3] F. Vanegas, D. Bratanov, K. Powell, J. Weiss, and F. Gonzalez, "A novel methodology for improving plant pest surveillance in vineyards and crops using UAV-based hyperspectral and spatial data," *Sensors*, vol. 18, no. 1, p. 260, Jan. 2018.

[4] J. Feng et al., "Class-aligned and class-balancing generative domain adaptation for hyperspectral image classification," *IEEE Trans. Geosci. Remote Sens.*, vol. 62, 2024, Art. no. 5509617.

[5] J. Feng, G. Bai, D. Li, X. Zhang, R. Shang, and L. Jiao, "MR-selection: A meta-reinforcement learning approach for zero-shot hyperspectral band selection," *IEEE Trans. Geosci. Remote Sens.*, vol. 61, 2023, Art. no. 5500320.

[6] M. Zhang, W. Li, X. Zhao, H. Liu, R. Tao, and Q. Du, "Morphological transformation and spatial-logical aggregation for tree species classification using hyperspectral imagery," *IEEE Trans. Geosci. Remote Sens.*, vol. 61, 2023, Art. no. 5501212.

[7] C. D. Abraham and J. Aravinth, "Fusion methods for hyperspectral image and LiDAR data at pixel-level," in *Proc. Int. Conf. Wireless Commun., Signal Process. Netw. (WiSPNET)*, Mar. 2018, pp. 1–3. Accessed: Dec. 11, 2023.

[8] M. Pedergnana, P. R. Marpu, M. D. Mura, J. A. Benediktsson, and L. Bruzzone, "Classification of remote sensing optical and LiDAR data using extended attribute profiles," *IEEE J. Sel. Topics Signal Process.*, vol. 6, no. 7, pp. 856–865, Nov. 2012.

[9] D. Hong, X. Wu, P. Ghamisi, J. Chanussot, N. Yokoya, and X. X. Zhu, "Invariant attribute profiles: A spatial-frequency joint feature extractor for hyperspectral image classification," *IEEE Trans. Geosci. Remote Sens.*, vol. 58, no. 6, pp. 3791–3808, Jun. 2020.

[10] Y. Zhang and S. Prasad, "Multisource geospatial data fusion via local joint sparse representation," *IEEE Trans. Geosci. Remote Sens.*, vol. 54, no. 6, pp. 3265–3276, Jun. 2016.

[11] B. Rasti, P. Ghamisi, J. Plaza, and A. Plaza, "Fusion of hyperspectral and LiDAR data using sparse and low-rank component analysis," *IEEE Trans. Geosci. Remote Sens.*, vol. 55, no. 11, pp. 6354–6365, Nov. 2017.

[12] D. Hong, J. Hu, J. Yao, J. Chanussot, and X. X. Zhu, "Multimodal remote sensing benchmark datasets for land cover classification with a shared and specific feature learning model," *ISPRS J. Photogramm. Remote Sens.*, vol. 178, pp. 68–80, Aug. 2021.

[13] M. A. Hearst, S. T. Dumais, E. Osuna, J. Platt, and B. Scholkopf, "Support vector machines," *IEEE Intell. Syst. Appl.*, vol. 13, no. 4, pp. 18–28, Aug. 1998.

[14] B. Bigdeli, F. Samadzadegan, and P. Reinartz, "A decision fusion method based on multiple support vector machine system for fusion of hyperspectral and LiDAR data," *Int. J. Image Data Fusion*, vol. 5, no. 3, pp. 196–209, Jul. 2014.

[15] Y. Zhong, Q. Cao, J. Zhao, A. Ma, B. Zhao, and L. Zhang, "Optimal decision fusion for urban land-use/land-cover classification based on adaptive differential evolution using hyperspectral and LiDAR data," *Remote Sens.*, vol. 9, no. 8, p. 868, Aug. 2017.

[16] J. Xia, N. Yokoya, and A. Iwasaki, "Fusion of hyperspectral and LiDAR data with a novel ensemble classifier," *IEEE Geosci. Remote Sens. Lett.*, vol. 15, no. 6, pp. 957–961, Apr. 2018.

[17] S. Li, W. Song, L. Fang, Y. Chen, P. Ghamisi, and J. A. Benediktsson, "Deep learning for hyperspectral image classification: An overview," *IEEE Trans. Geosci. Remote Sens.*, vol. 57, no. 9, pp. 6690–6709, Sep. 2019.

[18] Y. Gao et al., "Hyperspectral and multispectral classification for coastal wetland using depthwise feature interaction network," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, 2022, Art. no. 5512615.

[19] X. Xu, W. Li, Q. Ran, Q. Du, L. Gao, and B. Zhang, "Multisource remote sensing data classification based on convolutional neural network," *IEEE Trans. Geosci. Remote Sens.*, vol. 56, no. 2, pp. 937–949, Feb. 2018.

[20] R. Hang, Z. Li, P. Ghamisi, D. Hong, G. Xia, and Q. Liu, "Classification of hyperspectral and LiDAR data using coupled CNNs," *IEEE Trans. Geosci. Remote Sens.*, vol. 58, no. 7, pp. 4939–4950, Jul. 2020.

[21] S. Fang, K. Li, and Z. Li, "S^2ENet: Spatial–spectral cross-modal enhancement network for classification of hyperspectral and LiDAR data," *IEEE Geosci. Remote Sens. Lett.*, vol. 19, pp. 1–5, 2022.

[22] W. Dong, T. Zhang, J. Qu, S. Xiao, T. Zhang, and Y. Li, "Multibranch feature fusion network with self- and cross-guided attention for hyperspectral and LiDAR classification," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, 2022, Art. no. 5530612.

[23] S. Zhang, X. Meng, Q. Liu, G. Yang, and W. Sun, "Feature-decision level collaborative fusion network for hyperspectral and LiDAR classification," *Remote Sens.*, vol. 15, no. 17, p. 4148, Aug. 2023.

[24] W. Dong, T. Yang, J. Qu, T. Zhang, S. Xiao, and Y. Li, "Joint contextual representation model-informed interpretable network with dictionary aligning for hyperspectral and LiDAR classification," *IEEE Trans. Circuits Syst. Video Technol.*, vol. 33, no. 11, pp. 6804–6818, Nov. 2023.

[25] Y. Yang, J. Qu, W. Dong, T. Zhang, S. Xiao, and Y. Li, "TMCFN: Text-supervised multidimensional contrastive fusion network for hyperspectral and LiDAR classification," *IEEE Trans. Geosci. Remote Sens.*, vol. 62, 2024.

[26] M. Fei, J. Li, and H. Liu, "Visual tracking based on improved foreground detection and perceptual hashing," *Neurocomputing*, vol. 152, pp. 413–428, Mar. 2015.

[27] V. Monga and B. L. Evans, "Perceptual image hashing via feature points: Performance evaluation and tradeoffs," *IEEE Trans. Image Process.*, vol. 15, no. 11, pp. 3452–3465, Nov. 2006.

[28] G. Xu, Y. Wang, L. Wang, L. P. Soares, and C. H. Grohmann, "Feature-based constraint deep CNN method for mapping rainfall-induced landslides in remote regions with mountainous terrain: An application to Brazil," *IEEE J. Sel. Topics Appl. Earth Observ. Remote Sens.*, vol. 15, no. 1, pp. 2644–2659, May 2022.

[29] G. Huang, Z. Liu, L. Van Der Maaten, and K. Q. Weinberger, "Densely connected convolutional networks," in *Proc. IEEE Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jul. 2017, pp. 2261–2269, 2023.

[30] C. Debes et al., "Hyperspectral and LiDAR data fusion: Outcome of the 2013 GRSS data fusion contest," *IEEE J. Sel. Topics Appl. Earth Observ. Remote Sens.*, vol. 7, no. 6, pp. 2405–2418, Jun. 2014.

[31] P. Ghamisi, J. A. Benediktsson, and S. Phinn, "Land-cover classification using both hyperspectral and LiDAR data," *Int. J. Image Data Fusion*, vol. 6, no. 3, pp. 189–215, Jul. 2015.

[32] P. Gader, A. Zare, R. Close, J. Aitken, and G. Tuell, *Muufl Gulfport Hyperspectral and LiDAR Airborne Data Set*, document REP-2013-570, Gainesville, FL, USA, 2013.

[33] X. Du and A. Zare, *Technical Report: Scene Label Ground Truth Map for MUUFL Gulfport Data Set*. Gainesville, FL, USA: University of Florida, 2017.

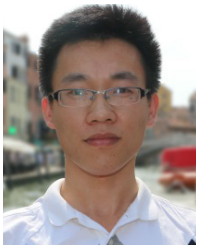
[34] F. Melgani and L. Bruzzone, "Classification of hyperspectral remote sensing images with support vector machines," *IEEE Trans. Geosci. Remote Sens.*, vol. 42, no. 8, pp. 1778–1790, Aug. 2004.

[35] J. Ham, Y. Chen, M. M. Crawford, and J. Ghosh, "Investigation of the random forest framework for classification of hyperspectral data," *IEEE Trans. Geosci. Remote Sens.*, vol. 43, no. 3, pp. 492–501, Mar. 2005.

[36] D. Hong, J. Chanussot, N. Yokoya, J. Kang, and X. X. Zhu, "Learning-shared cross-modality representation using multispectral-LiDAR and hyperspectral data," *IEEE Geosci. Remote Sens. Lett.*, vol. 17, no. 8, pp. 1470–1474, Aug. 2020.

[37] X. Wang, Y. Feng, R. Song, Z. Mu, and C. Song, "Multi-attentive hierarchical dense fusion net for fusion classification of hyperspectral and LiDAR data," *Inf. Fusion*, vol. 82, pp. 1–18, Jun. 2022.

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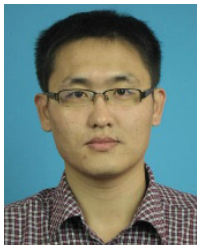
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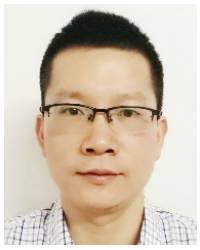
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